GA-03 | Basics of Computer Applications

RRB JE CBT-2 (CEN 05/2025) — General Abilities | Weightage: 10 Questions

> **Exam angle:** RRB asks direct, factual, application-level questions — shortcut keys, MS Office functions, memory types, security terms, open-source concepts. This chapter covers every sub-topic seen in actual 2025 CBT-2 papers, followed by a PYQ Corner, 25 practice MCQs, and rapid-revision tables.

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1. Computer Fundamentals & Architecture

1.1 What is a Computer?

A computer is an electronic device that accepts data (**input**), processes it according to stored instructions (**processing**), produces results (**output**), and stores data for future use (**storage**). This is the **IPOS cycle**: Input → Process → Output → Storage.

1.2 Generations of Computers

Generation	Period	Technology	Examples
First	1940–56	Vacuum tubes	ENIAC, UNIVAC
Second	1956–63	Transistors	IBM 1401, IBM 7094
Third	1964–71	Integrated Circuits (IC)	IBM 360
Fourth	1971–present	Microprocessors (VLSI)	IBM PC, Apple II
Fifth	Present & beyond	AI / ULSI / Parallel processing	AI systems, robotics

- **Father of Computer:** Charles Babbage (Analytical Engine).
- **Father of Modern Computer / Stored-program concept:** John von Neumann.
- **First electronic general-purpose computer:** ENIAC.
- **India's first supercomputer:** PARAM 8000 (C-DAC, 1991).

1.3 The Central Processing Unit (CPU)

The CPU is the **"brain of the computer"**. It has three main parts:

Component	Full Form	Function
ALU	Arithmetic Logic Unit	Performs all arithmetic (+, −, ×, ÷) and logical (AND, OR, NOT, comparisons) operations
CU	Control Unit	Directs and coordinates all operations; fetches, decodes and executes instructions; does **not** process data itself
Registers	—	Small, fastest storage locations inside the CPU that hold data/instructions currently being processed

Important registers:

Register	Role
Accumulator (AC)	Holds intermediate arithmetic/logic results
Program Counter (PC)	Holds address of the **next** instruction
Instruction Register (IR)	Holds the **current** instruction being executed
Memory Address Register (MAR)	Holds address of memory location to be accessed
Memory Data/Buffer Register (MDR/MBR)	Holds data being transferred to/from memory

- **Instruction cycle:** Fetch → Decode → Execute → Store.
- **Clock speed** is measured in **Hertz (GHz)** — cycles per second.
- **Word length** — number of bits the CPU processes at a time (32-bit, 64-bit).

1.4 Data Measurement Units

Unit	Value
1 Bit	Binary digit (0 or 1) — smallest unit
1 Nibble	4 bits
1 Byte	8 bits
1 KB	1024 Bytes
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB
1 PB	1024 TB

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2. Memory Hierarchy ★ (Frequently asked)

Memory is arranged from **fastest & smallest** (top) to **slowest & largest** (bottom):

Registers → Cache → Main memory (RAM) → Secondary storage (HDD/SSD) → Tertiary (tape/optical)

2.1 Primary Memory (Main Memory)

(a) RAM — Random Access Memory — volatile (contents lost when power is off); holds programs/data currently in use.

Feature	SRAM (Static RAM)	DRAM (Dynamic RAM)
Storage element	Flip-flops (6 transistors)	Capacitor + transistor
Refreshing	**NOT required ★**	**Required** periodically (charge leaks)
Speed	Faster	Slower
Cost	Costlier	Cheaper
Density	Lower (bulkier per bit)	Higher (more bits per chip)
Power	Lower at idle	Higher (refresh cycles)
Used in	**Cache memory**	**Main memory (system RAM)**

> **2025 CBT-2 asked:** Which memory does *not* need refreshing? → **SRAM** (it uses flip-flops, so data stays as long as power is on; DRAM stores charge on capacitors that leak and must be refreshed thousands of times per second).

(b) ROM — Read Only Memory — non-volatile; holds firmware/boot instructions (BIOS).

ROM Type	Full Form	Key Point
PROM	Programmable ROM	Written **once** by user; cannot be erased
EPROM	Erasable Programmable ROM	Erased by **ultraviolet (UV) light**, reprogrammable
EEPROM	Electrically Erasable PROM	Erased **electrically**, byte-by-byte
Flash memory	—	A type of EEPROM erased in blocks; used in pen drives, SSDs, memory cards

2.2 Cache Memory

- Very fast **SRAM-based** memory between CPU and RAM.
- Stores frequently used data/instructions to reduce average access time.
- Levels: **L1** (inside CPU core, fastest, smallest), **L2**, **L3** (shared, largest, slowest of the three).

2.3 Secondary Storage

Device	Type	Notes
Hard Disk Drive (HDD)	Magnetic	Rotating platters; capacity in TB
Solid State Drive (SSD)	Flash (electronic)	No moving parts; much faster than HDD
CD / DVD / Blu-ray	Optical	CD ≈ 700 MB, DVD ≈ 4.7 GB, Blu-ray ≈ 25 GB/layer
Pen drive / Memory card	Flash	Portable, uses USB
Magnetic tape	Magnetic	Sequential access; backups/archives

- **Virtual memory:** part of the hard disk used as an extension of RAM by the operating system.
- **Buffer:** temporary holding area for data being transferred between devices.

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3. Input & Output Devices

Input Devices	Output Devices	Both (I/O)
Keyboard, Mouse, Scanner, Joystick, Light pen, Trackball, Microphone, Webcam, Barcode reader, OMR, OCR, MICR (bank cheques), Biometric sensor, Digitizer/Graphics tablet	Monitor (VDU), Printer, Plotter, Speaker, Projector, Headphones	Touchscreen, Modem, Network card, Digital camera, Headset with mic, Fax machine, Multifunction printer

Printers:

Type	Category	Notes
Dot matrix	Impact	Strikes ribbon; noisy; can print carbon copies
Daisy wheel	Impact	Letter-quality, character printer
Inkjet	Non-impact	Sprays ink droplets
Laser	Non-impact	Uses toner + drum; fast, high quality
Plotter	—	Draws large engineering drawings/graphs

- Monitor sharpness → **resolution** (pixels); printer quality → **DPI** (dots per inch); printer speed → **PPM** (pages per minute).
- **OMR** = Optical Mark Recognition (exam answer sheets); **OCR** = Optical Character Recognition; **MICR** = Magnetic Ink Character Recognition (cheque numbers).

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4. Software Concepts

Software = set of programs/instructions. Two broad types:

Type	Description	Examples
System software	Controls & manages hardware; platform for other software	Operating systems, device drivers, utilities (antivirus, disk cleanup), language translators
Application software	Performs specific user tasks	MS Word, Excel, browsers, Tally, Photoshop

Language translators:

Translator	Function
Compiler	Translates entire high-level program to machine code at once (C, C++)
Interpreter	Translates and executes line by line (Python, BASIC)
Assembler	Translates assembly language to machine code

Software licence types:

Term	Meaning
Proprietary	Source code closed; paid licence (Windows, MS Office)
Open-source	**Source code freely available to view, modify and redistribute** (Linux, LibreOffice, Firefox, Android/AOSP)
Freeware	Free to use, but source code not open (Adobe Reader)
Shareware	Free trial, then payment (WinRAR)

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5. Operating Systems ★

An **Operating System (OS)** is system software that acts as an **interface between the user and hardware** and manages all resources.

Functions of an OS:

1. **Process management** — scheduling CPU among programs.
2. **Memory management** — allocating/deallocating RAM.

3. **File management** — creating, organizing, protecting files/folders.
4. **Device management** — via device drivers; I/O control.
5. **Security** — user authentication, access control.
6. **User interface** — CLI (command line) or GUI (graphical).

Popular Operating Systems:

OS	Developer	Type	Key facts
Windows	Microsoft	Proprietary, GUI	Windows 11 latest desktop version; file system NTFS
Linux	Linus Torvalds (kernel, 1991)	**Open-source ★**	Free; distributions: Ubuntu, Fedora, Debian, Red Hat, BOSS (India, C-DAC)
macOS	Apple	Proprietary	Runs only on Apple Macs; UNIX-based
Android	Google	Open-source (Linux-kernel based)	World's most used mobile OS
iOS	Apple	Proprietary	iPhone/iPad
MS-DOS	Microsoft	CLI, single-user	Historic; commands like DIR, COPY
UNIX	AT&T Bell Labs	Multi-user	Ken Thompson & Dennis Ritchie

> **2025 CBT-2 asked:** Which OS is open-source? → **Linux**. Its source code is publicly available and can be modified/redistributed under the GNU GPL. Windows and macOS are proprietary.

- **Booting** = starting the computer and loading the OS. Cold boot = from power-off; warm boot = restart (Ctrl+Alt+Del).
- **Kernel** = core of the OS, always resident in memory.
- **Multitasking** = several programs at once; **multiprocessing** = multiple CPUs; **multi-user** = many users share one system; **real-time OS** = instant response (used in railways signalling, embedded systems).

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6. MS Word (Word Processing) ★

- Default file extension: **.docx** (older: .doc). A Word file is called a **Document**.
- Default view: Print Layout. Ruler/gridlines under **View** tab; margins/orientation under **Layout**; header-footer, page number, tables, pictures under **Insert**; spelling & grammar under **Review** (shortcut **F7**).

6.1 Function Keys F1–F12 (Windows/MS Office) ★

Key	Function
F1	Help
F2	Rename selected file/folder (Windows); **in MS Word: move selected text/graphic** — select text → press F2 → click new location → press Enter ★
F3	Search (Windows); Shift+F3 in Word = change case
F4	Repeat last action (Word); Alt+F4 = close window/program
F5	Refresh (browser/Windows); Find & Replace "Go To" (Word); start slideshow (PowerPoint)
F6	Move cursor between screen panes / to address bar (browsers)
F7	Spelling & Grammar check (Word)
F8	Extend selection (Word); Safe Mode boot menu (Windows startup)
F9	Update fields (Word); refresh (Outlook send/receive)
F10	Activate menu bar; Shift+F10 = right-click menu
F11	Full-screen mode (browsers); new chart sheet (Excel)
F12	Save As (Word/Excel/PowerPoint)

> **2025 CBT-2 asked:** Which key moves selected text in MS Word? → **F2** (select the text, press F2, place cursor at destination, press Enter).

6.2 Common MS Word Shortcuts

Shortcut	Action	Shortcut	Action
Ctrl+N	New document	Ctrl+B	Bold
Ctrl+O	Open	Ctrl+I	Italic
Ctrl+S	Save	Ctrl+U	Underline
Ctrl+P	Print	Ctrl+L / E / R / J	Align left / centre / right / justify
Ctrl+C / X / V	Copy / Cut / Paste	Ctrl+K	Insert hyperlink
Ctrl+Z / Y	Undo / Redo	Ctrl+F	Find
Ctrl+A	Select all	Ctrl+H	Replace
Ctrl+Home / End	Start / end of document	Ctrl+Enter	Page break
Ctrl+Shift+C	Copy formatting (Format Painter)	Ctrl+] / [	Increase / decrease font size by 1 pt
Ctrl+=	Subscript	Ctrl+Shift+=	Superscript

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7. MS Excel (Spreadsheet) ★

- File extension: **.xlsx** (older: .xls). File = **Workbook**; each page = **Worksheet**; intersection of row & column = **Cell** (address like A1, B5).

- Excel columns: A, B, C…; rows: 1, 2, 3…. Modern Excel: 1,048,576 rows × 16,384 columns (last column **XFD**).
- Every formula begins with **=**.

7.1 Cell References

Type	Example	Behaviour when copied
Relative	=A1+B1	Adjusts with position
Absolute	=$A$1	Never changes ($ locks row & column); toggle with **F4**
Mixed	=$A1 or =A$1	One part locked

7.2 Common Functions ★

Function	Syntax	Purpose
SUM	=SUM(A1:A10)	Adds numbers
AVERAGE	=AVERAGE(A1:A10)	Arithmetic mean
COUNT	=COUNT(A1:A10)	Counts cells containing **numbers**
COUNTA	=COUNTA(A1:A10)	Counts non-empty cells
MAX / MIN	=MAX(range)	Largest / smallest value
IF	=IF(A1>40,"Pass","Fail")	Logical test with two outcomes
VLOOKUP	=VLOOKUP(value, table, col_index, FALSE)	Looks up a value in the **first column** of a table and returns a value from another column in the same row
HLOOKUP	=HLOOKUP(...)	Horizontal lookup (first row)
TEXT	**=TEXT(value, "format")**	**Converts a numeric value to text in a specified number format** ★ e.g. =TEXT(1234.5,"₹#,##0.00") → "₹1,234.50"; =TEXT(TODAY(),"DD-MM-YYYY")
CONCATENATE / &	=A1&B1	Joins text strings
TODAY() / NOW()	—	Current date / date-time
ROUND	=ROUND(A1,2)	Rounds to given decimals

> **2025 CBT-2 asked:** Function that converts a value to text in a given format? → **TEXT()**. It formats numbers/dates as text strings, e.g. =TEXT(0.25,"0%") returns "25%".

7.3 Charts & Other Features

- Chart types: Column, Bar, Line, **Pie** (parts of a whole), Scatter (correlation), Area. Insert via **Insert tab → Charts**. **F11** creates an instant chart on a new sheet.
- **Fill handle** (small square at cell corner) drags formulas/series.
- **Pivot Table** summarizes large data; **Conditional formatting** colours cells by rules; **Freeze Panes** (View tab) keeps headings visible; **Sort & Filter** under Data tab.
- Excel shortcut specials: Ctrl+; = today's date, Ctrl+Shift+; = current time, Alt+= = AutoSum.

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8. MS PowerPoint (Presentations) ★

- File extension: **.pptx** (older: .ppt); a file is a **Presentation** made of **Slides**. Slide show file: .ppsx.
- **Views:** Normal, Outline, **Slide Sorter** (thumbnails of all slides), Notes Page, Reading, **Slide Show** (F5 from start; Shift+F5 from current slide).
- **Slide Master** (View tab) controls formatting of all slides at once.
- **Design tab → Themes / Templates:** ready-made combinations of colours, fonts, effects and background styles applied to the whole presentation. ★
- **Transitions** = effects between slides; **Animations** = effects on objects within a slide.
- New slide: Ctrl+M. End show: Esc.

> **2025 CBT-2 asked:** Where do you apply a design template/theme in PowerPoint? → the **Design tab** (choose a theme/template; older versions: Format → Slide Design).

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9. File Extensions — One Table ★

Extension	File Type	Extension	File Type
.docx / .doc	Word document	.txt	Plain text
.xlsx / .xls	Excel workbook	.rtf	Rich text
.pptx / .ppt	PowerPoint	.pdf	Portable Document Format (Adobe)
.jpg / .jpeg, .png, .gif, .bmp	Images	.html / .htm	Web page
.mp3, .wav	Audio	.mp4, .avi, .mkv	Video
.zip, .rar	Compressed archive	.exe	Windows executable
.accdb / .mdb	MS Access database	.csv	Comma-separated values
.c / .cpp / .py / .java	Source code	.dll	Dynamic link library
.tmp	Temporary file	.bak	Backup file

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10. Internet Basics

Term	Meaning
Internet	Global network of networks; grew out of **ARPANET** (1969, USA)
WWW	World Wide Web — system of interlinked hypertext documents; invented by **Tim Berners-Lee (1989, CERN)**
Web browser	Software to access web pages — Chrome, Firefox, Edge, Safari, Opera; first popular browser: Mosaic
Search engine	Finds information on the web — Google, Bing, DuckDuckGo, Yahoo
URL	Uniform Resource Locator — address of a web page (protocol://domain/path)
HTTP / HTTPS	HyperText Transfer Protocol / Secure (encrypted with SSL/TLS)
HTML	HyperText Markup Language — language of web pages
ISP	Internet Service Provider (Jio, Airtel, BSNL)
Homepage	First page of a website
Download / Upload	Receiving from / sending to a remote server
Cookies	Small files a website stores on your computer to remember you

E-mail:

- Format: username@domain (e.g. rrbje@gmail.com). Invented by **Ray Tomlinson** (chose the @ sign).
- **CC** = Carbon Copy (all recipients visible); **BCC** = Blind Carbon Copy (hidden recipients).
- Protocols: **SMTP** (sending), **POP3 / IMAP** (receiving). Attachment = file sent with mail. Spam = unsolicited bulk mail.

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11. Computer Networking (One Page)

Network	Coverage	Example
PAN	Personal, ~10 m	Bluetooth devices
LAN	Local Area Network — building/campus	Office network, computer lab
MAN	Metropolitan — a city	Cable TV network
WAN	Wide Area Network — country/world	**The Internet** (largest WAN)

Devices & terms:

Item	Role
Modem	Modulator-Demodulator; converts digital↔analog signals
Router	Connects different networks; routes data packets
Switch / Hub	Connect devices within a LAN (switch is intelligent, hub broadcasts)
IP address	Unique numerical address of a device (IPv4: 32-bit, e.g. 192.168.1.1; IPv6: 128-bit)
DNS	Domain Name System — translates domain names (google.com) into IP addresses ("phonebook of the Internet")
MAC address	Physical hardware address of a network card (48-bit)
Bandwidth	Data-carrying capacity, in bps/Mbps
TCP/IP	The basic protocol suite of the Internet
Wi-Fi	Wireless LAN (IEEE 802.11); Bluetooth: short-range (IEEE 802.15)
Topology	Layout of a network — bus, star (most common), ring, mesh, tree
Server / Client	Provider of services / requester of services

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12. Computer Security ★★ (Guaranteed question zone)

12.1 Firewall ★

- A **firewall** is a security system (hardware, software, or both) that **monitors and controls incoming and outgoing network traffic** based on predefined security rules, and **blocks unauthorised access** to or from a private network.
- It sits **between a trusted internal network and an untrusted external network (Internet)** — like a security gate.
- A firewall does **not** primarily remove viruses (that is antivirus software); it filters traffic.

> **2025 CBT-2 asked:** Function of a firewall? → It **monitors/filters network traffic and prevents unauthorised access** to a computer or private network.

12.2 Malware & Threats

Threat	Definition
Virus	Malicious code that attaches itself to files/programs and spreads **when the host file is executed**; needs human action
Worm	Standalone malware that **self-replicates and spreads across networks without human action** or a host file
Trojan horse	Malware disguised as legitimate/useful software; does **not** self-replicate; opens backdoors
Phishing	Fraudulent emails/websites impersonating trusted entities to **steal passwords, banking and personal data**
Ransomware	Encrypts the victim's files and **demands payment (ransom)** for the decryption key (e.g. WannaCry, 2017)
Spyware	Secretly monitors user activity and sends data to attackers
Adware	Forces unwanted advertisements
Keylogger	Records every keystroke to capture passwords
Hacking	Unauthorised access to systems; "ethical hackers" test security legally
Spoofing	Faking identity (IP/email) to appear trusted
DoS/DDoS attack	Flooding a server with traffic to make it unavailable

12.3 Protective Measures

- **Antivirus software** — detects, quarantines and removes malware; must be **updated regularly** (examples: Norton, McAfee, Quick Heal, Kaspersky).
- **Strong password** — at least 8–12 characters mixing **uppercase, lowercase, digits and symbols**; never use name/DOB; change periodically; use different passwords per account; enable **two-factor authentication (2FA/OTP)**.
- **HTTPS + padlock icon** — indicates an **encrypted (SSL/TLS)** connection; always check before online payments.
- **Encryption** — converting readable data (plaintext) into coded form (ciphertext); decryption reverses it.
- **Backup** — regular copies of data guard against ransomware and disk failure.
- **CERT-In** — Indian Computer Emergency Response Team, national nodal agency for cyber security incidents (under MeitY). IT Act 2000 governs cyber law in India.

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13. Websites & E-Governance in India

Initiative	Purpose
Digital India (2015)	Umbrella programme for digital empowerment
UMANG app	Single app for many government services
DigiLocker	Cloud storage of official documents (certificates, DL, Aadhaar)
UPI (NPCI)	Unified Payments Interface — instant bank transfers (BHIM app)
Aadhaar (UIDAI)	12-digit biometric identity number
IRCTC (irctc.co.in)	Online railway ticket booking
PMO India, india.gov.in	National portal of India
e-NAM	National Agriculture Market
CoWIN / eSanjeevani	Vaccination / telemedicine platforms
GeM	Government e-Marketplace for public procurement
Domain suffixes	.gov.in (Indian govt), .com (commercial), .org (organisation), .edu (education), .in (India country code)

14. Latest Tech — One-Liners

Technology	One-liner
Artificial Intelligence (AI)	Machines simulating human intelligence (learning, reasoning); e.g. ChatGPT (OpenAI), chess engines; subfields: Machine Learning, Deep Learning
Cloud computing	On-demand computing services (storage, servers, software) over the Internet — "pay as you go"; models: IaaS, PaaS, SaaS; providers: AWS, Azure, Google Cloud
IoT (Internet of Things)	Everyday devices (bulbs, ACs, watches) embedded with sensors and connected to the Internet
Big Data	Extremely large data sets analysed for patterns (Volume, Velocity, Variety)
Blockchain	Distributed, tamper-proof digital ledger; basis of Bitcoin cryptocurrency
5G	Fifth-generation mobile network; launched in India Oct 2022
Virtual/Augmented Reality	VR: fully simulated environment; AR: digital overlay on real world
Quantum computing	Uses qubits; can solve certain problems far faster than classical computers
Machine Learning	Systems that learn from data without explicit programming
Robotics/Drones	Automated machines; drones = unmanned aerial vehicles (UAV)

15. PYQ Corner — Actual RRB JE CBT-2 2025 Questions Solved

Q1. What is the primary function of a firewall in a computer network?

Answer: It monitors and controls incoming and outgoing network traffic based on security rules and **blocks unauthorised access** to the system/network. It acts as a barrier between a trusted internal network and the untrusted Internet.

Q2. In MS Word, which function key is used to move selected text or graphics to a new location?

Answer: F2. Select the text → press F2 → click at the destination → press **Enter**. (F4 repeats last action, F7 = spell check, F12 = Save As.)

Q3. Which MS Excel function converts a numeric value into text in a specified number format?

Answer: TEXT() — syntax =TEXT(value, format_text), e.g. =TEXT(1234.567,"0.00") → "1234.57". Useful for displaying dates/currency as formatted text.

Q4. Which of the following is an open-source operating system?

Answer: Linux. Its source code is freely available under the GNU General Public License and can be modified and redistributed. Windows and macOS are proprietary. (Android is also open-source, built on the Linux kernel.)

Q5. Which type of RAM does not require periodic refreshing?

Answer: SRAM (Static RAM) — it stores data in flip-flops which retain data as long as power is supplied. **DRAM** stores data as charge on capacitors that leak, so it must be refreshed continuously; hence "dynamic".

Q6. In MS PowerPoint, a design template/theme is applied from which tab?

Answer: The **Design tab** — it offers Themes (predefined sets of colours, fonts, effects and backgrounds) that can be applied to all slides or selected slides.

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16. Practice MCQs (25) — Answers at the End

1. The "brain" of the computer is the: (a) RAM (b) CPU (c) ROM (d) Monitor
2. Which unit performs arithmetic and logical operations? (a) CU (b) ALU (c) MU (d) Register
3. 1 Byte equals: (a) 4 bits (b) 8 bits (c) 16 bits (d) 1024 bits
4. Cache memory is made of: (a) DRAM (b) SRAM (c) EPROM (d) Magnetic disk
5. Which memory is volatile? (a) ROM (b) Hard disk (c) RAM (d) Flash drive
6. EPROM is erased using: (a) Electric pulse (b) Ultraviolet light (c) Magnet (d) Heat
7. Which is NOT an input device? (a) Scanner (b) Joystick (c) Plotter (d) OMR
8. MICR is mainly used in: (a) Railways (b) Banks (c) Hospitals (d) Schools
9. Which translator converts a program line by line? (a) Compiler (b) Assembler (c) Interpreter (d) Linker
10. Which of these is application software? (a) Windows 11 (b) Linux (c) MS Excel (d) Device driver
11. The core part of an operating system is called the: (a) Shell (b) Kernel (c) BIOS (d) Registry
12. Ubuntu and Fedora are distributions of: (a) Windows (b) macOS (c) Linux (d) DOS
13. In MS Word, Ctrl+H opens: (a) Help (b) Hyperlink (c) Find and Replace (d) Header
14. F7 in MS Word is used for: (a) Save As (b) Spell check (c) Print (d) Thesaurus only
15. In Excel, =$B$2 is an example of: (a) Relative reference (b) Absolute reference (c) Mixed reference (d) Circular reference
16. Which Excel function counts only numeric cells? (a) COUNTA (b) COUNT (c) SUM (d) COUNTIF
17. VLOOKUP searches for a value in the: (a) Last column (b) First row (c) First column (d) Any column
18. Shift+F5 in PowerPoint: (a) Starts show from first slide (b) Starts show from current slide (c) Closes show (d) Prints slides

19. The view that shows thumbnails of all slides is: (a) Normal (b) Reading (c) Slide Sorter (d) Notes Page
20. .csv stands for: (a) Computer system values (b) Comma-separated values (c) Column-sorted values (d) Central stored values
21. WWW was invented by: (a) Bill Gates (b) Tim Berners-Lee (c) Vint Cerf (d) Ray Tomlinson
22. DNS translates: (a) IP to MAC (b) Domain names to IP addresses (c) HTML to HTTP (d) Bits to bytes
23. Malware that self-replicates over networks without a host file is a: (a) Virus (b) Trojan (c) Worm (d) Adware
24. An attack that encrypts files and demands payment is: (a) Phishing (b) Spoofing (c) Ransomware (d) Keylogging
25. HTTPS indicates: (a) High-speed connection (b) Encrypted secure connection (c) Government site (d) Free site

Answers: 1-b, 2-b, 3-b, 4-b, 5-c, 6-b, 7-c, 8-b, 9-c, 10-c, 11-b, 12-c, 13-c, 14-b, 15-b, 16-b, 17-c, 18-b, 19-c, 20-b, 21-b, 22-b, 23-c, 24-c, 25-b

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17. Rapid-Revision Tables

17.1 Ten Shortcuts You Must Not Forget

Shortcut	Action	Shortcut	Action
Ctrl+C / X / V	Copy / Cut / Paste	F2	Rename (Windows) / Move text (Word)
Ctrl+Z / Y	Undo / Redo	F5	Refresh / Slide show
Ctrl+S / F12	Save / Save As	F7	Spell check
Ctrl+P	Print	Alt+F4	Close program
Ctrl+A	Select all	Ctrl+Alt+Del	Task options / warm boot

17.2 Memory in 30 Seconds

Point	Remember
Fastest memory	Registers → then Cache (SRAM)
Needs refresh	DRAM (capacitor) — main RAM
No refresh	SRAM (flip-flop) — cache
Non-volatile primary	ROM (BIOS); EPROM = UV erase; EEPROM = electric erase
Virtual memory	Hard-disk space used as RAM

17.3 Security Terms in One Line Each

Term	One line
Firewall	Filters network traffic; blocks unauthorised access
Virus	Needs host file + human action
Worm	Self-spreading, no host needed
Trojan	Disguised as useful software
Phishing	Fake mail/site steals credentials
Ransomware	Encrypts data, demands money
Antivirus	Detects and removes malware
HTTPS	SSL/TLS-encrypted browsing

17.4 Office Suite Snapshot

Application	File called	Extension	Star fact
MS Word	Document	.docx	F2 moves text; F7 spell check
MS Excel	Workbook	.xlsx	TEXT() formats value as text; $ = absolute reference
MS PowerPoint	Presentation	.pptx	Design tab → themes/templates; F5 slide show

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End of GA-03. Next: GA-04 Basics of Environment & Pollution Control.